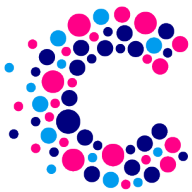
Multiple Comparisons

Luca Porcu & Chandra Chilamakuri (Bioinformatics Core)

Cancer Research UK Cambridge Institute, University of Cambridge

26 October 2026, 09:30–12:30

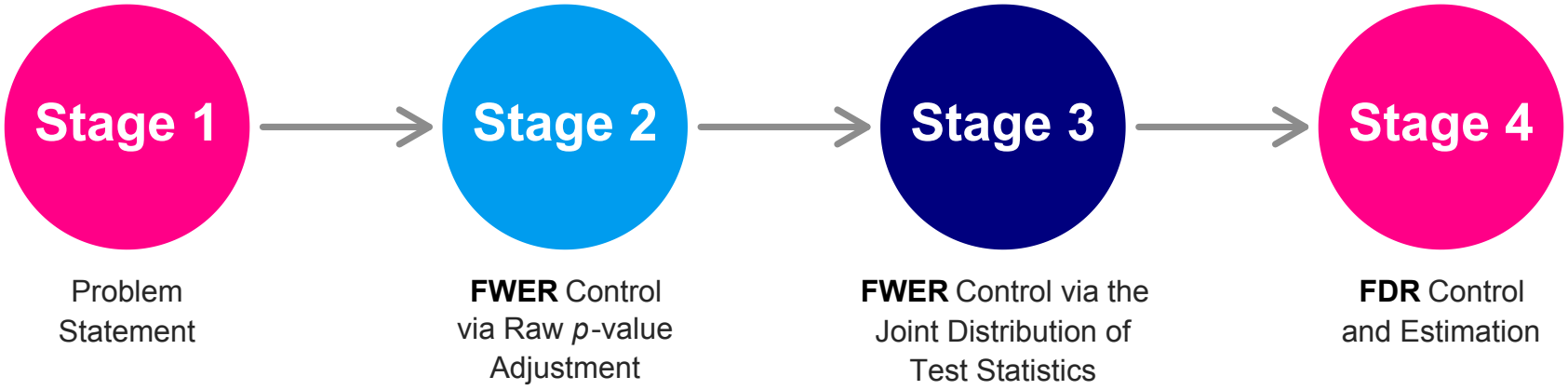


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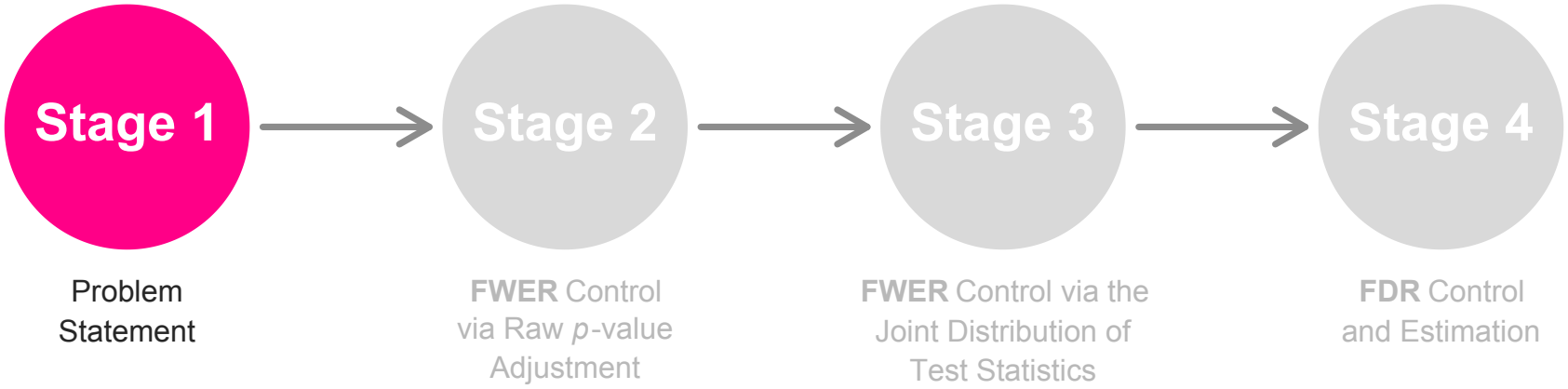
Roadmap



FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Roadmap



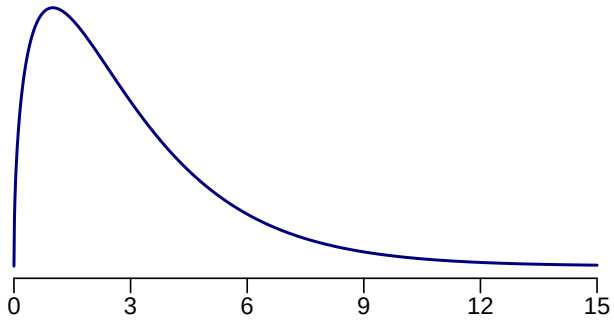
FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Statistical concepts used for data-driven decisions

1 **Population** (i.e. statistical model) $\mathcal{P}_\theta$ → 2 A **hypothesis** (H) is a statement about $\mathcal{P}_\theta$

e.g. $\theta = 5$
$\theta \leq 10$
$\theta_1 = \theta_2$ (i.e. Pop. 1 = Pop. 2)



The underlying population gives rise to the data

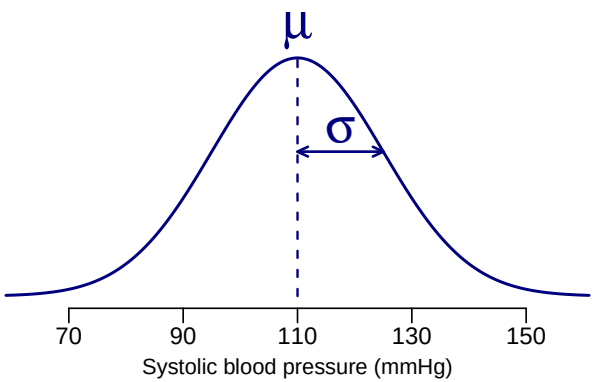
$d(x_1, \ldots, x_m)$

3 **Data** $(x_1, \ldots, x_m)$ → 4 $D = \{$ TRUE, FALSE $\}$

A decision function $d(x_1, \ldots, x_m)$ is a function whose domain is the set of possible values of the data and whose range is the set of possible decisions (decision space D)

A hypothesis about μ

1 **Population**: UK adults, $\mathcal{P}_{\mu,\sigma}$



The underlying population gives rise to the data

2 A **hypothesis** (H) is a statement about $\mathcal{P}_{\mu,\sigma}$

H: $\mu = 110$ mmHg
(i.e. the mean systolic blood pressure of UK adults is 110 mmHg)

3 **Data**: $(132, 128, 139, 125, 141, 134)$

$d(x_1, \ldots, x_m)$

4 $D = \{$ TRUE, FALSE $\}$

The decision function $d(x_1, \ldots, x_m)$ is **Student's *t*-test** at significance level α (e.g. 0.05).
It maps the observed data to $D = \{\text{TRUE, FALSE}\}$

Null hypothesis (H_0) and statistical errors

- We can never prove that a hypothesis (H_0) is TRUE
- We can **reject** H_0 (declare it FALSE), accepting a **controlled probability** α of rejecting it when it is actually TRUE

Hence, we want to control the probability of false positive statements (i.e. avoiding false statements about the population)

H_0	Not Rejected	Rejected
TRUE	$1-\alpha$	α
FALSE	β	$1-\beta$

$\alpha = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ TRUE})$: **Type I error rate** (false positive, significance level)

$\beta = \mathrm{P}(\text{fail to reject } H_0 \mid H_0 \text{ FALSE})$: **Type II error rate** (false negative)

$1-\beta = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ FALSE})$: **Statistical power** (true positive)

Hypothesis tests as decision functions

1. A hypothesis (H_0) makes a statement about a population (i.e. statistical model)



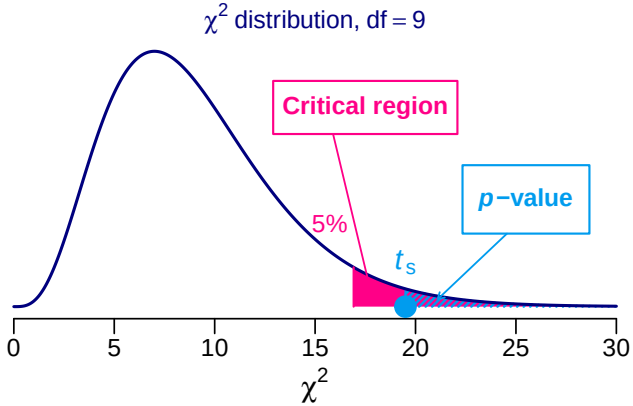
2. To test H_0, a sample from the population is collected



3. Data $(x_1, \ldots, x_m)$ are collected. A **test statistic t_s**, which is a function of the sample data, is calculated

$$t_s = t(x_1, \ldots, x_m)$$

4. Under H_0, the distribution of the test statistic t is known. The calculated test statistic t_s is located on this distribution



$d(t_s) =$ FALSE if t_s falls in the critical region

Student's *t*-test

1. H_0: the mean systolic blood pressure of UK adults is $\mu_0 = 110$ mmHg



2. To test H_0, a sample from the population is collected

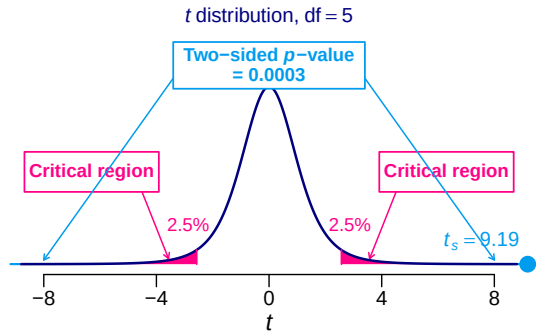


3. Data **(132, 128, 139, 125, 141, 134)** are collected. A test statistic t_s **(Student's *t*)**, which is a function of the sample data, is calculated

$$t_s = \frac{\bar{x} - \mu_0}{s/\sqrt{m}} = \frac{133.17 - 110}{6.18/\sqrt{6}}$$

$$= \frac{23.17}{2.52} = \mathbf{9.19}, \quad \text{df} = m - 1 = 5$$

4. Under H_0, the distribution of the test statistic t is known. The calculated test statistic t_s is located on this distribution



$d(t_s) =$ FALSE because t_s falls in the critical region

The multiple testing problem

Single null hypothesis

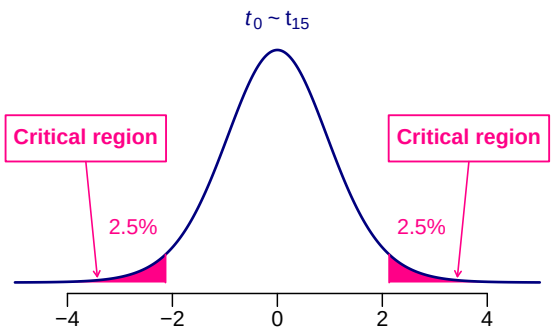
Let H_0 be a null hypothesis

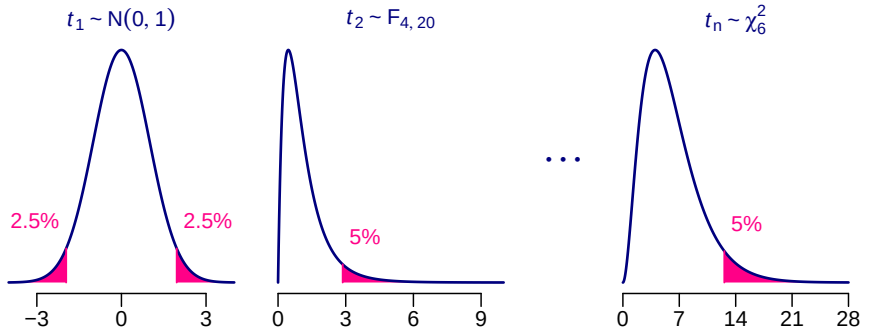
Family of null hypotheses

Let $\mathcal{F} = \{H_1, H_2, \dots, H_n\}$ be a family of null hypotheses

We apply a statistical test to H_0 and to each $H_i \in \mathcal{F}$

Under each null hypothesis, the distribution of its test statistic is known:





If H_0 is TRUE, the probability of a false positive statement is 5%

If all the H_i, $i = 1, 2, \dots, n$, are TRUE, the probability of at least one false positive statement (FWER) is $\geq 5\%$ and depends on the joint distribution of the test statistics t_i

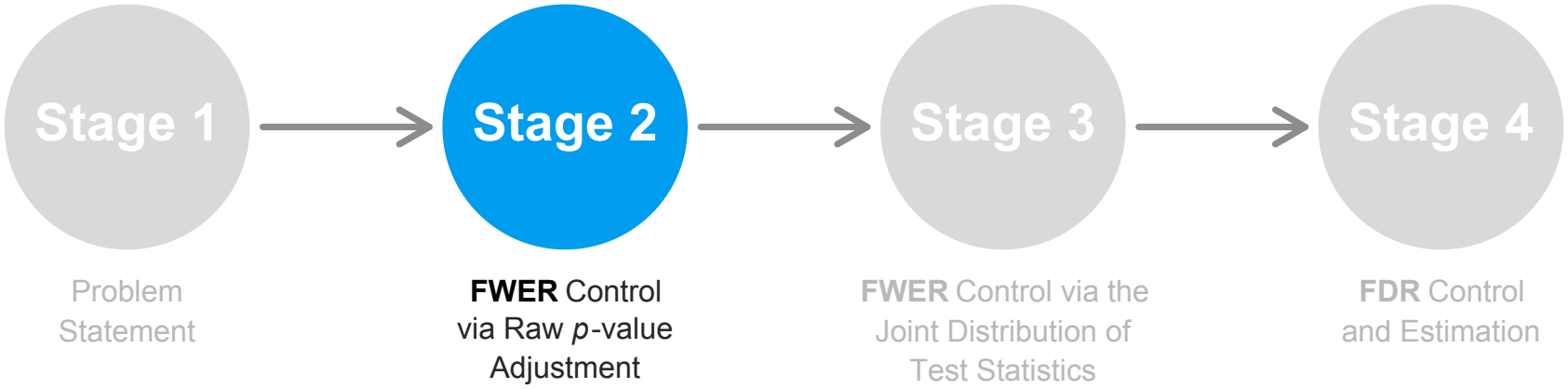
The multiple testing problem



If we go on testing long enough, we will inevitably find something that is “statistically significant”

A *p*-value controls the error rate of *its own* test, not that of the whole family

Roadmap



FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Family-Wise Error Rate (FWER)

Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses, and let m_0 ($0 \leq m_0 \leq n$) denote the number of TRUE **null hypotheses**.

$\mathcal{F}$	Not Rejected	Rejected	Total
TRUE	U	V	m_0
FALSE	T	S	$n - m_0$
Total	$n - R$	R	n

- Because the test statistics follow a probability distribution, the counts U, V, T, S, $n - R$ and R also follow probability distributions (i.e. they are random variables). The quantities $n - R$ and R are observable
- The counts m_0, $n - m_0$ and n are **fixed**, not random: n is known, whereas m_0 and $n - m_0$ are **unknown**

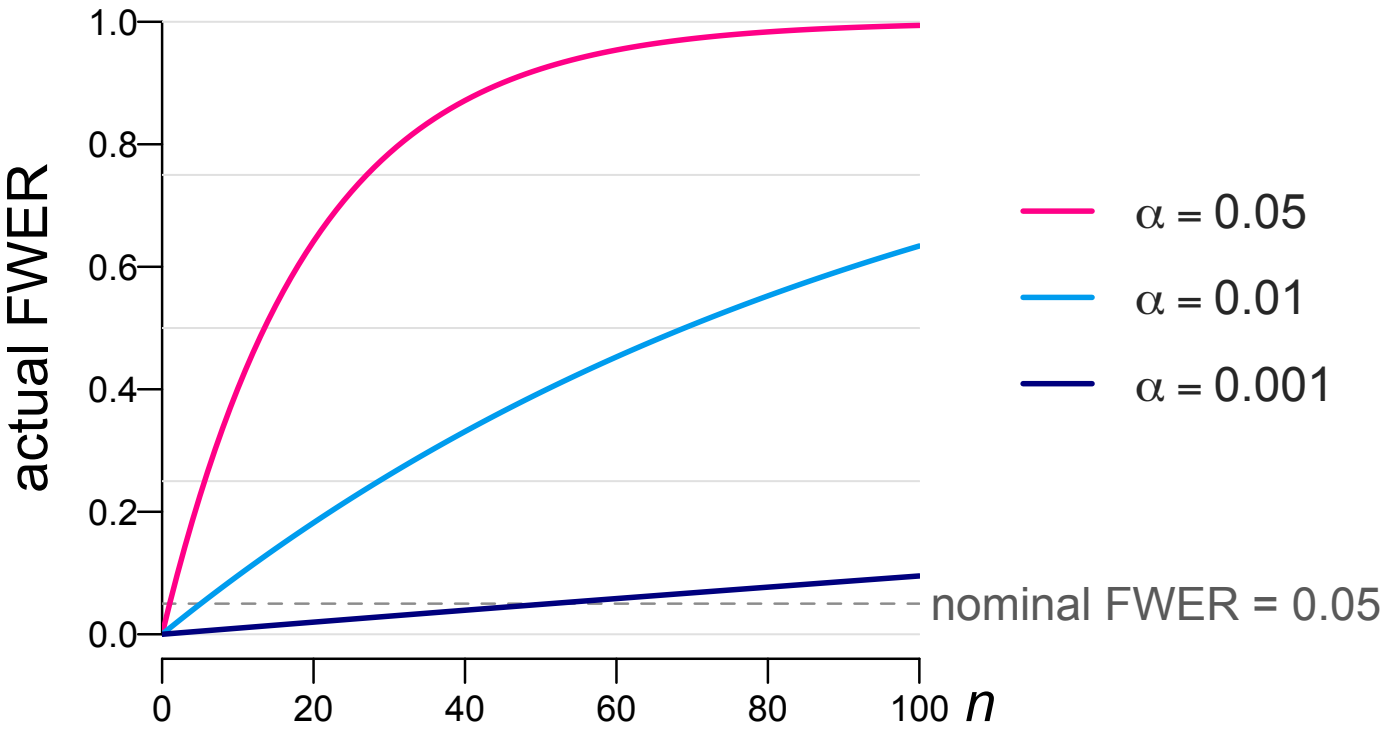
FWER $:=$ **P(** V **> 0)**

We want to control the FWER **whatever m_0 is** and whichever nulls are TRUE.

We accept only a small probability α (e.g. 0.05) of at least one false positive!!!

FWER and raw *p*-values

- Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of TRUE null hypotheses: this is the **global null hypothesis**, H_0^{global}
- Each hypothesis is tested at a pre-specified significance level α
- Assuming the *n* tests are **independent**, FWER = $1 - (1 - \alpha)^n$



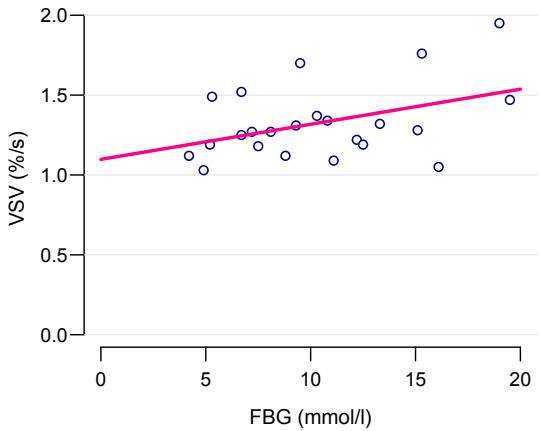
FWER and raw *p*-values

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

H_1: $\mu_{VSV} = 0$ at FBG = 0

H_2: no linear association between FBG and VSV (i.e. slope = 0)

A simple linear model is used to test both hypotheses

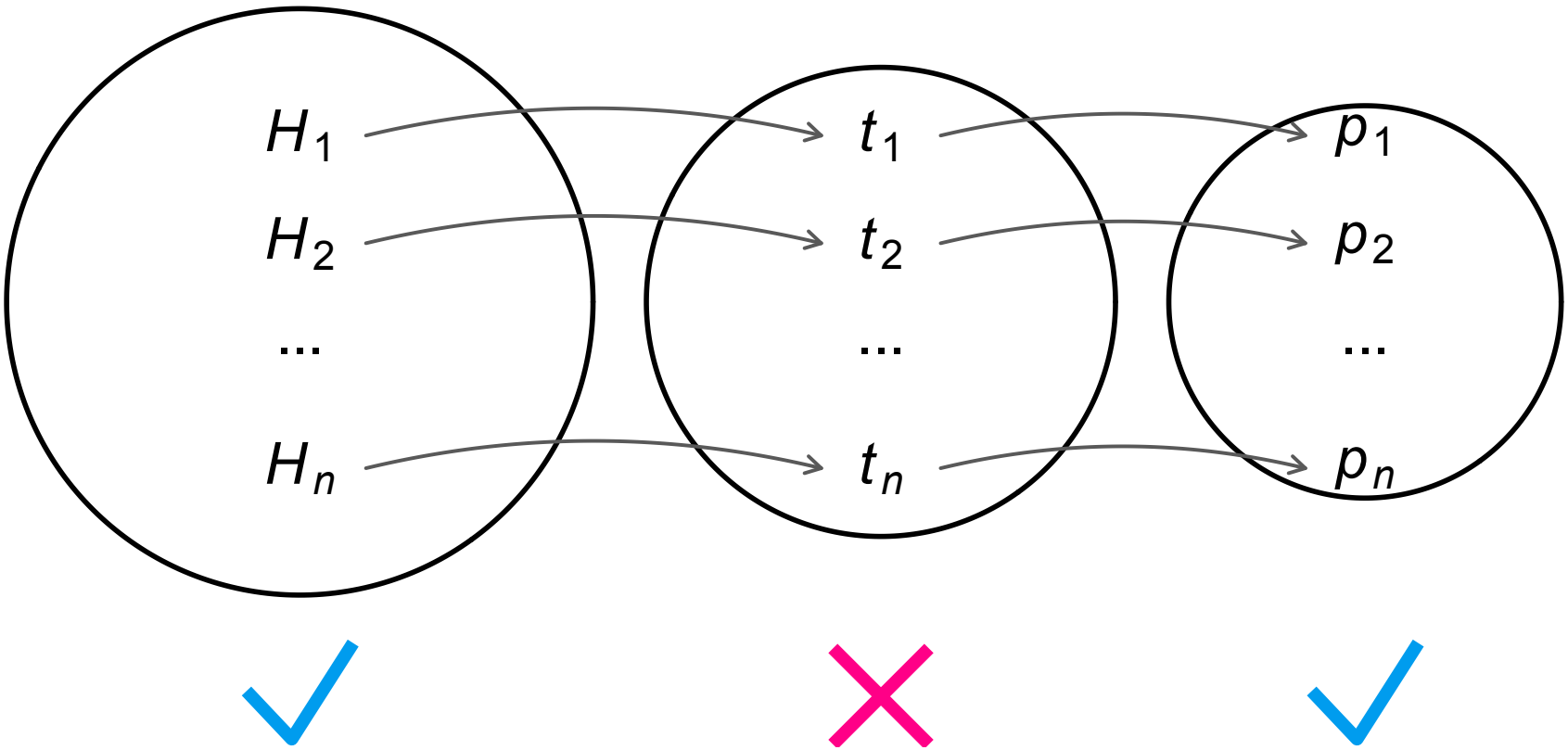



```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Is this a false positive result?

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Adjustment of raw *p*-values



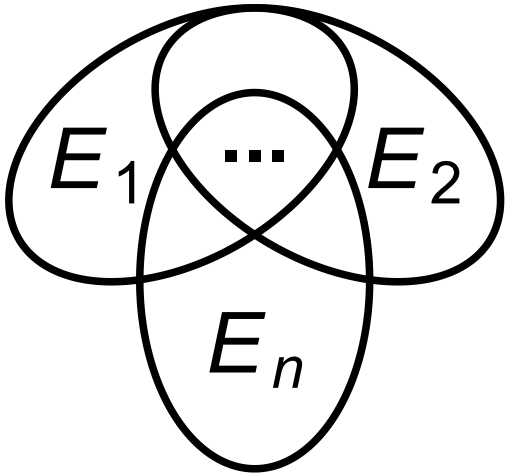
Here we do not use the multivariate distribution of the test statistics $(t_1, t_2, \ldots, t_n)$, working instead with the marginal *p*-values (p_i) of the individual null hypotheses (H_i).

Bonferroni correction

Boole's inequality

Let $E_1, E_2, \ldots, E_n$ be events.

$$\mathrm{P}\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} \mathrm{P}(E_i)$$



Boole's inequality is applied in hypothesis testing

E_i := {Under a TRUE H_i, the test statistic t_i falls inside the **critical region**}

$$\text{FWER} = \mathrm{P}\left(\bigcup_{i \in K_0} E_i\right)$$ (K_0 = set of the m_0 TRUE null hypotheses)

- We want FWER $\leq \alpha$ (e.g. 0.05)
- Each test is run at level $\frac{\alpha}{n}$, so $\mathrm{P}(E_i) = \frac{\alpha}{n}$ for every $i \in K_0$
- Hence, FWER $\leq \sum_{i \in K_0} \mathrm{P}(E_i) = \frac{\alpha}{n} \times m_0 \leq \alpha$

Bonferroni correction

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

H_1: $\mu_{\text{VSV}} = 0$ at FBG = 0

H_2: no linear association between FBG and VSV (i.e. slope = 0)

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Adjusted *p*-value

<0.001

0.096

Bonferroni does not reject H_2 at $\alpha = 0.05$

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Correction rule

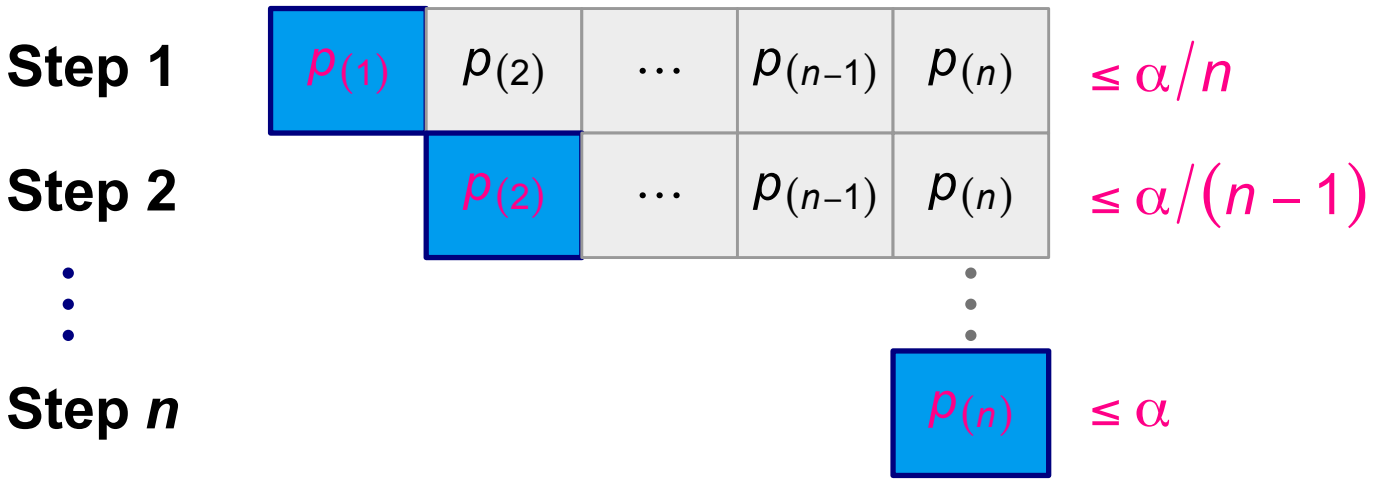
Raw *p*-values	**Adjusted *p*-values**
p_1	$n \times p_1$
p_2	$n \times p_2$
$\vdots$	$\vdots$
p_n	$n \times p_n$

Note: adjusted *p*-values must satisfy $\tilde{p}_i \leq 1$, hence $\tilde{p}_i = \min(1,\ n \times p_i)$

Holm–Bonferroni correction

We apply Boole's inequality in a step-down procedure

- Order the n p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$, with corresponding hypotheses $H_{(1)}, H_{(2)}, \ldots, H_{(n)}$



- Stop at the first non-rejection: the remaining hypotheses are all retained

It controls the FWER at the α level, but it is less conservative than Bonferroni because the threshold **grows** after each rejection ($\frac{\alpha}{n}, \frac{\alpha}{n-1}, \frac{\alpha}{n-2}, \ldots$), whereas Bonferroni uses $\frac{\alpha}{n}$ throughout.

Holm–Bonferroni correction

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

H_1: $\mu_{\text{VSV}} = 0$ at FBG = 0

H_2: no linear association between FBG and VSV (i.e. slope = 0)

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Adjusted *p*-value

<0.001

0.048

Holm rejects H_2 at $\alpha = 0.05$

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Correction rule

Order the n p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$

Raw *p*-values	**Adjusted *p*-values**
$p_{(1)}$	$n \times p_{(1)}$
$p_{(2)}$	$(n-1) \times p_{(2)}$
$\vdots$	$\vdots$
$p_{(n)}$	$1 \times p_{(n)}$

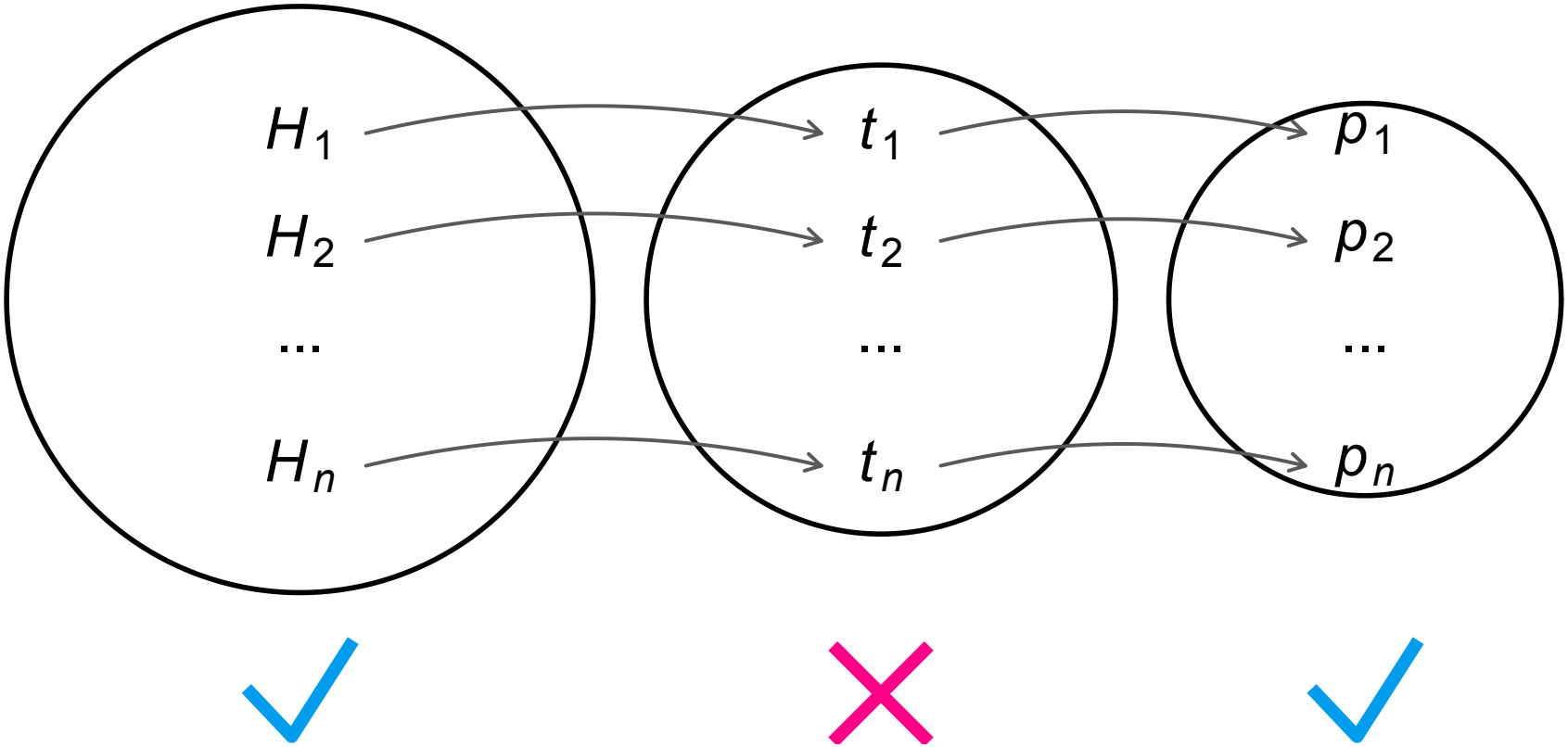
Note: adjusted p-values must satisfy

$$\tilde{p}_{(i)} \leq 1 \text{ and } \tilde{p}_{(i)} \leq \tilde{p}_{(i+1)}$$

General rule:

$$\tilde{p}_{(i)} = \max\left(\tilde{p}_{(i-1)},\ \min\left(1,\ (n-i+1) \times p_{(i)}\right)\right)$$

Assumptions on test statistic relationships



The corrections differ in what they assume about the **dependence** among the test statistics ($t_1, t_2, \ldots, t_n$): Bonferroni and Holm require **no assumption** about it, while Šidák assumes **independence**.

Šidák correction

- Let $H_1, H_2, \ldots, H_n$ be hypotheses with **independent** test statistics $t_1, t_2, \ldots, t_n$
- For each H_i, $i = 1, 2, \ldots, n$, set the significance level to $\alpha_{\text{Šidák}}$. Under H_0^{global},

$$\text{FWER} = 1 - (1 - \alpha_{\text{Šidák}})^n$$

- Set the FWER equal to α. Solving the previous equation for $\alpha_{\text{Šidák}}$, we obtain the individual significance level for each hypothesis H_i:

$$\alpha_{\text{Šidák}} = \mathbf{1} - (\mathbf{1} - \alpha)^{\frac{\mathbf{1}}{n}}$$

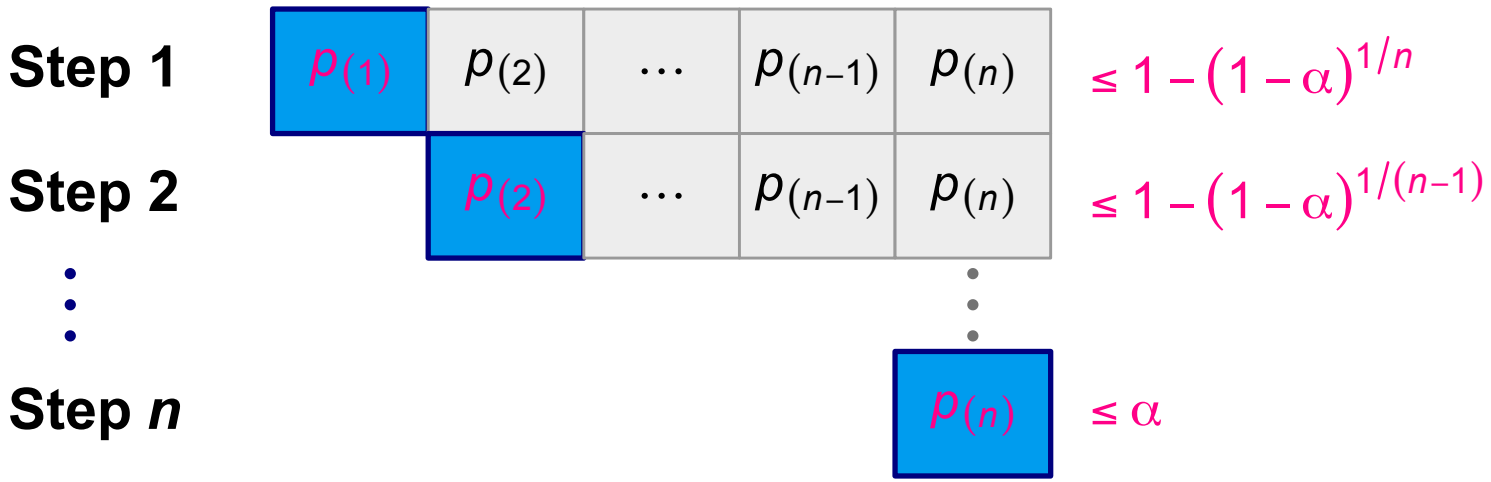
- Reject H_i if $p_i \leq \alpha_{\text{Šidák}}$, or equivalently if the adjusted p-value $\tilde{p}_i = 1 - (1 - p_i)^n \leq \alpha$

The Šidák correction is valid and, since $1 - (1 - \alpha)^{1/n} > \alpha/n$ for $n \geq 2$, it is slightly more powerful than the Bonferroni correction

Holm–Šidák correction

We apply the Šidák correction in a step-down procedure

- Order the n p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(n)}$, with corresponding hypotheses $H_{(1)}, H_{(2)}, \ldots, H_{(n)}$ (**independent** test statistics)

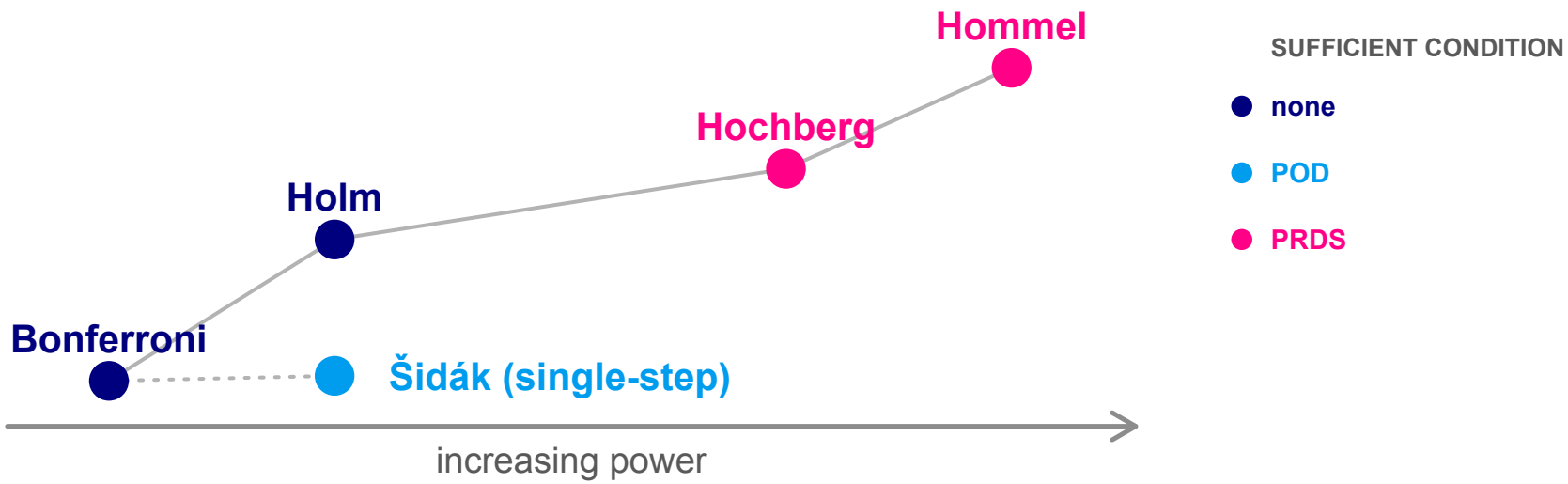


- Stop at the first non-rejection: the remaining hypotheses are all retained

It controls the FWER at the α level under independence, and it is slightly less conservative than Holm–Bonferroni since $1 - (1-\alpha)^{1/j} \geq \alpha/j,\ j = 1, 2, \ldots, n$

Which correction is the most powerful?

All five procedures **control** the FWER at α, each under its own condition. They differ in **how much power** they gain over Bonferroni, and in **what they must assume** to do so.



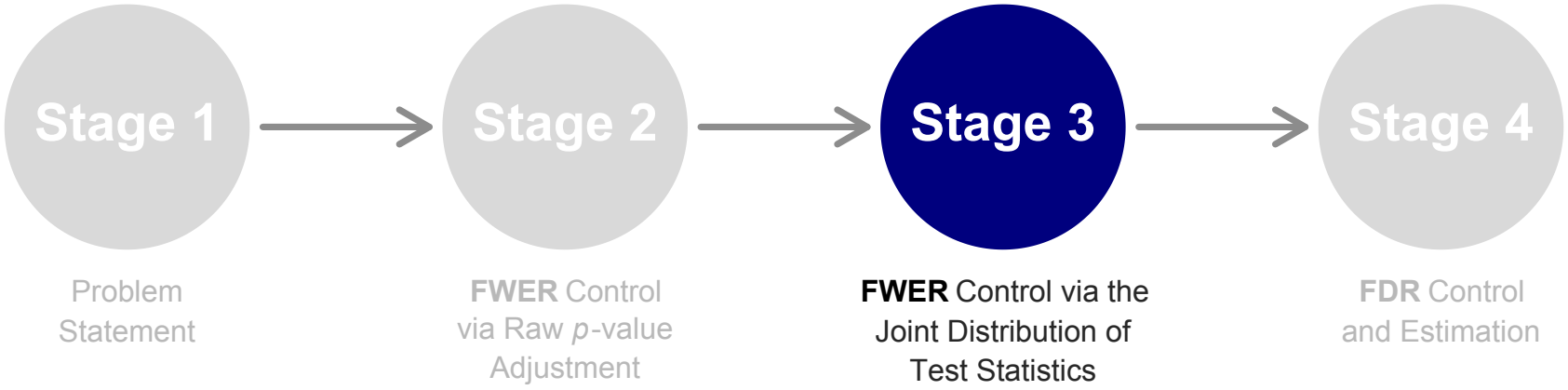
- **Šidák (single-step)** is **not uniformly more or less powerful** than **Holm**, **Hochberg** and **Hommel**: each can reject a hypothesis the other does not
- **Šidák**, **Hochberg** and **Hommel** all **control the FWER** when the test statistics are **independent**
- **POD** and **PRDS** are both **positive dependence** conditions (PRDS is the stronger one): one test being significant **never makes the others less likely** to be significant

Simulations and practicals

Shiny web applications

https://bioinformatics.cruk.cam.ac.uk/stats/shinystats/

Roadmap



FWER: Family-Wise Error Rate

FDR: False Discovery Rate

Joint distribution of the test statistics

Ventricular shortening velocity (VSV, %/s) and fasting blood glucose (FBG, mmol/l) were measured in 24 type 1 diabetic patients (23 with complete data). The following two null hypotheses are tested:

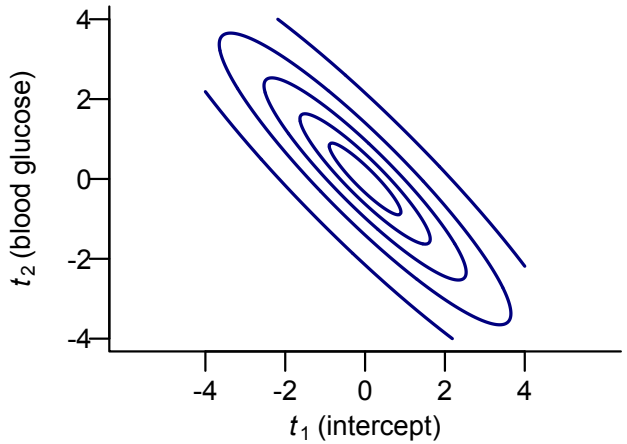
H_1: $\mu_{\text{VSV}} = 0$ at FBG $= 0$

H_2: no linear association between FBG and VSV (i.e. slope $= 0$)

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.098    0.117      9.345   <0.001
## blood.glucose 0.022    0.010      2.101   0.048
```

Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

Under H_0^{global}, we know the joint distribution of the test statistics



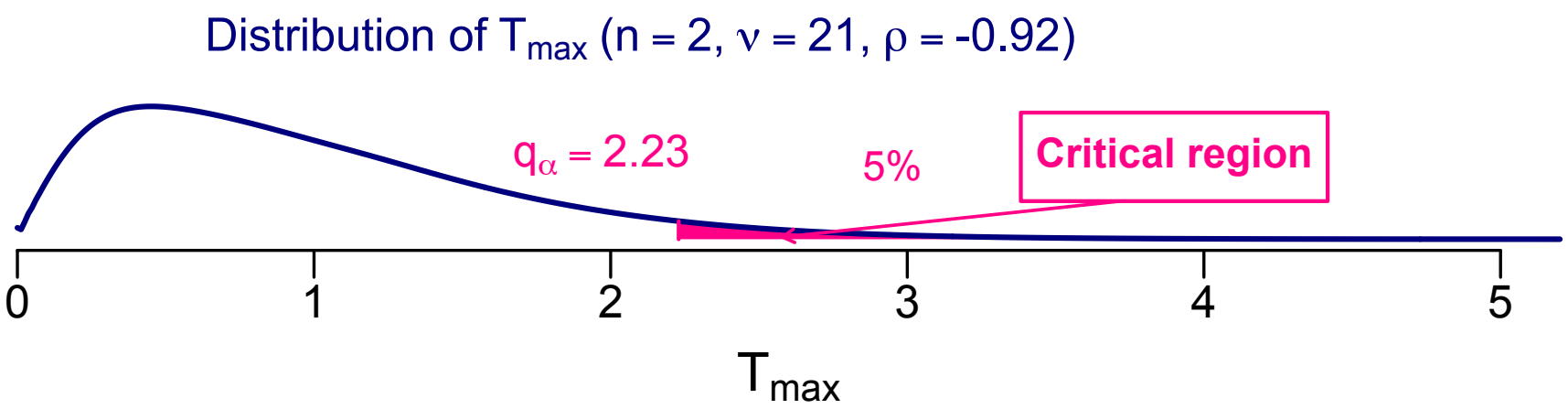
Bivariate Student's t distribution under H_0^{global} ($\nu = 21$, $\rho = -0.92$)

The max-*t* statistical principle

- Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses
- Let $\mathcal{D}$ be the joint distribution of the individual test statistics $t_1, t_2, \ldots, t_n$ under H_0^{global} (i.e. all hypotheses are TRUE). As in our example, the tests are two-sided: larger values of $|t_i|$ favour the rejection of H_i
- Consider the maximum of the individual test statistics t_i:

$$T_{\max} = \max_{i=1,2,\ldots,n} |t_i| \qquad (\text{one-sided tests: } T_{\max} = \max_i t_i)$$

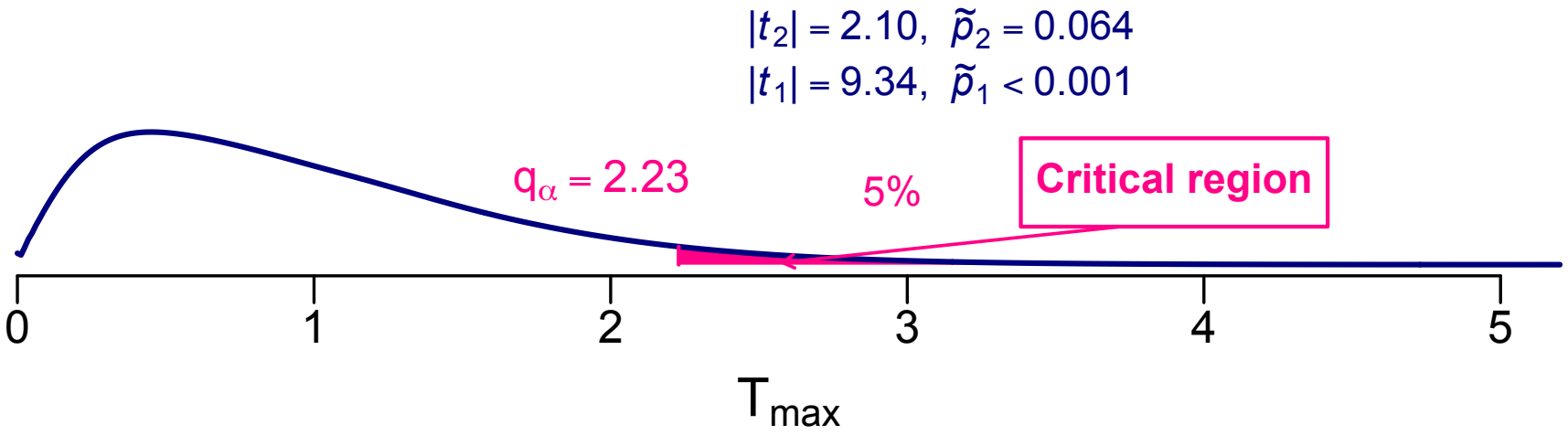
The joint distribution $\mathcal{D}$ determines the distribution of $T_{\max}$ under H_0^{global}:



Data: `thuesen` dataset (`ISwR` package v2.0-12, R 4.5.1)

The max-*t* statistical principle

- Consider the positions of the statistics $|t_1|, |t_2|, \ldots, |t_n|$:



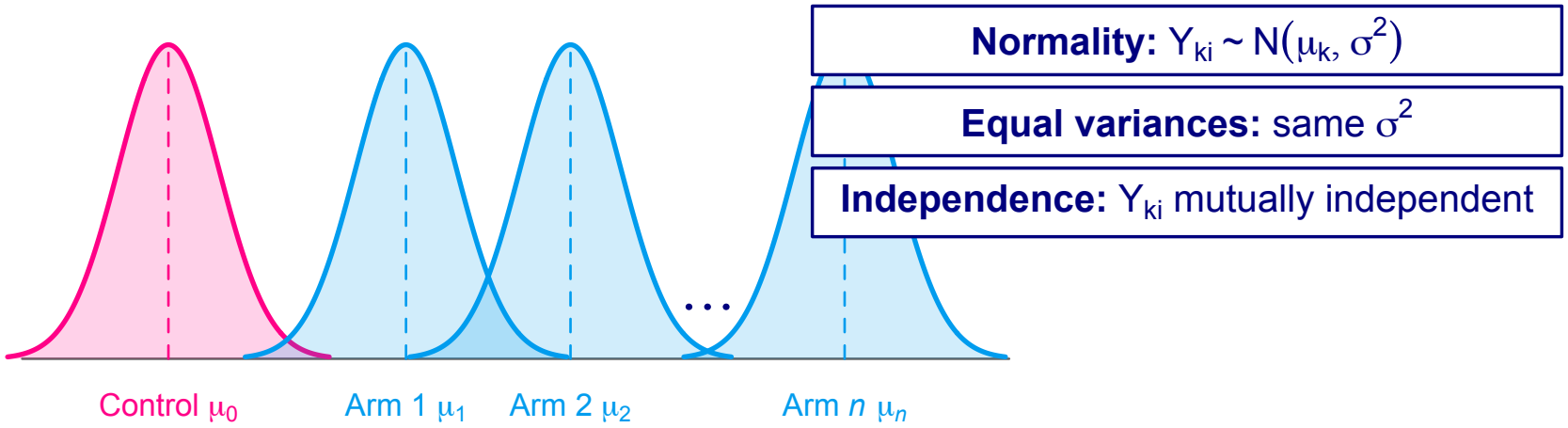
- H_1 is **rejected**: $|t_1| = 9.34 > q_\alpha = 2.23$, adjusted $\tilde{p}_1 < 0.001$
  H_2 is **not rejected**: $|t_2| = 2.10 < q_\alpha = 2.23$, adjusted $\tilde{p}_2 = 0.064 > 0.05$
  (single-step max-*t*: adjusted $\tilde{p}_2 = 0.064$; step-down max-*t*: adjusted $\tilde{p}_2 = 0.048$)

Subset pivotality condition: take **any** subset K of the hypotheses. Their statistics $\{t_i : i \in K\}$ must behave the same way when only the hypotheses in K are TRUE as they do when **all** of them are TRUE

Under the subset pivotality condition, the FWER is controlled (FWER $\leq \alpha$**)**

Dunnett's test

- **Statistical model**: one-way analysis of variance (ANOVA)



- **Hypotheses**: $H_k : \mu_k = \mu_0, k = 1, \ldots, n$

  For each of the n arms, the arm mean equals the control mean μ_0

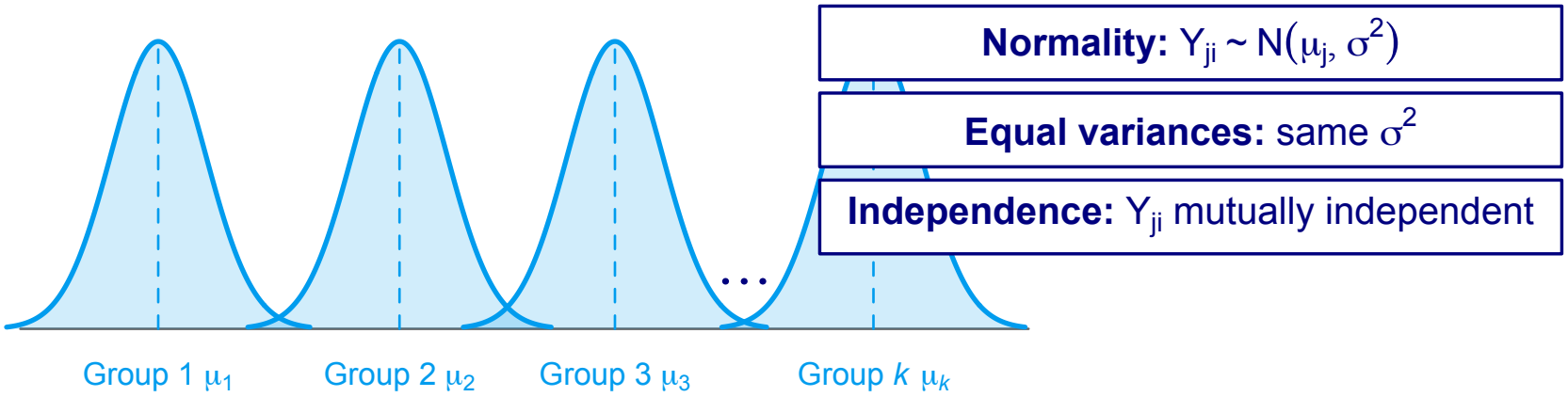
- Each t_k follows a Student's t distribution with $N - (n + 1)$ degrees of freedom under H_k

  $\mathcal{D}$ is a multivariate Student's t distribution under H_0^{global}

  ↳ **The max-t principle is applied to $\mathcal{D}$**

Tukey's test

- **Statistical model**: one-way analysis of variance (ANOVA)



- **Hypotheses**: $H_{j\ell} : \mu_j = \mu_\ell, 1 \leq j < \ell \leq k$

  For each of the $n = k(k-1)/2$ pairs, the two group means are equal

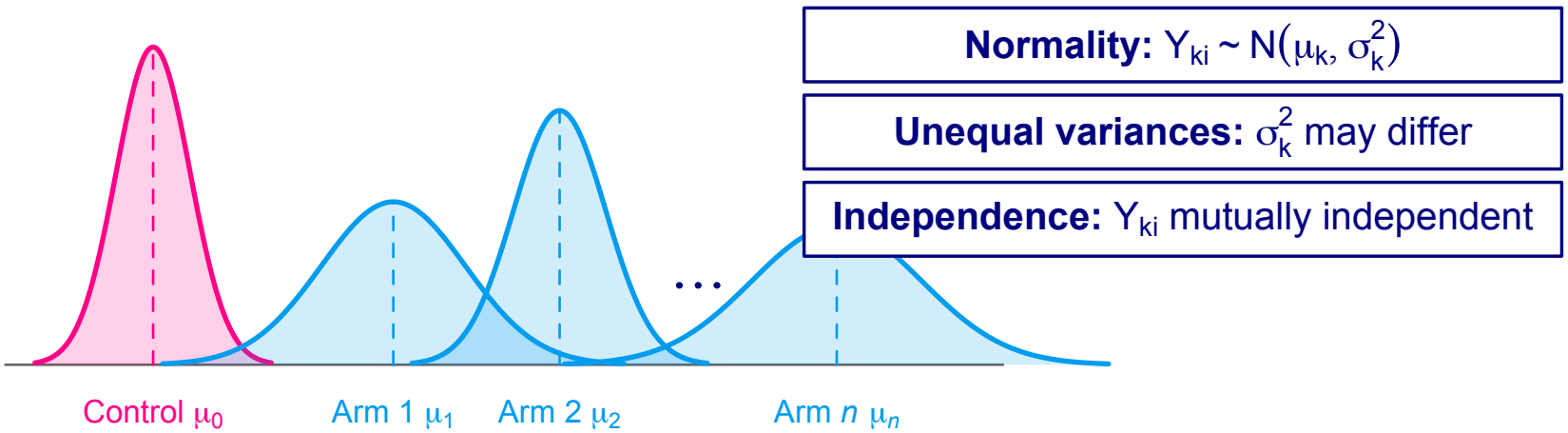
- Each $t_{j\ell}$ follows a Student's t distribution with $N - k$ degrees of freedom under $H_{j\ell}$

  $\mathcal{D}$ is a multivariate Student's t distribution under H_0^{global}

  → **The max-t principle is applied to $\mathcal{D}$**

Dunnett's test with unequal variances (Hasler–Hothorn)

- **Statistical model**: one-way ANOVA **without** the equal-variance assumption



- **Hypotheses**: $H_k : \mu_k = \mu_0, k = 1, \ldots, n$

  For each of the n arms, the arm mean equals the control mean μ_0

- Each t_k (Welch) follows **approximately** a Student's t distribution under H_k

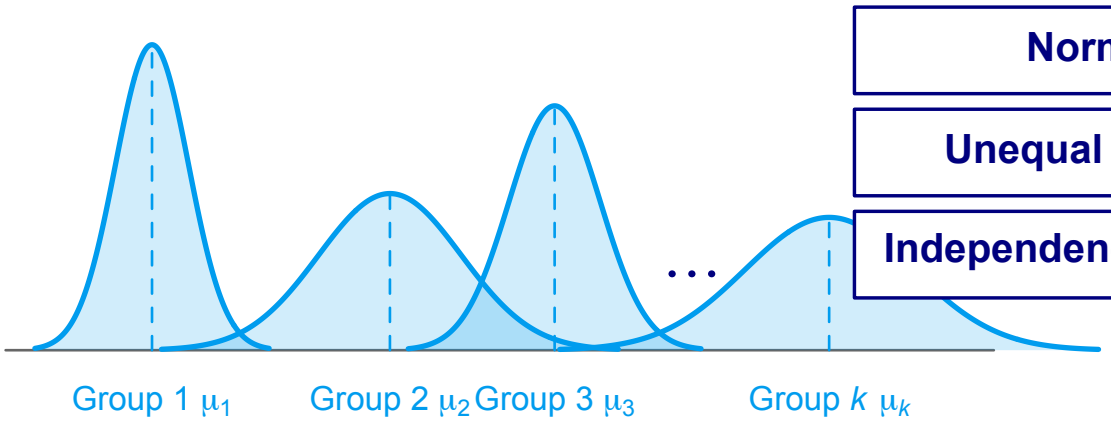
  $\mathcal{D}$ is **approximately** a multivariate Student's t distribution under H_0^{global}

  ↳ **The max-t principle is applied to $\mathcal{D}$**

Tamhane's T2 and **Dunnett's T3** treat $\mathcal{D}$ **marginally** (each statistic on its own, not jointly)

Tukey's test with unequal variances (Hasler–Hothorn)

- **Statistical model**: one-way ANOVA **without** the equal-variance assumption



Normality: $Y_{ji} \sim N(\mu_j, \sigma_j^2)$

Unequal variances: σ_j^2 may differ

Independence: Y_{ji} mutually independent

- **Hypotheses**: $H_{j\ell} : \mu_j = \mu_\ell, 1 \le j < \ell \le k$

  For each of the $n = k(k-1)/2$ pairs, the two group means are equal

- Each $t_{j\ell}$ (Welch) follows **approximately** a Student's t distribution under $H_{j\ell}$

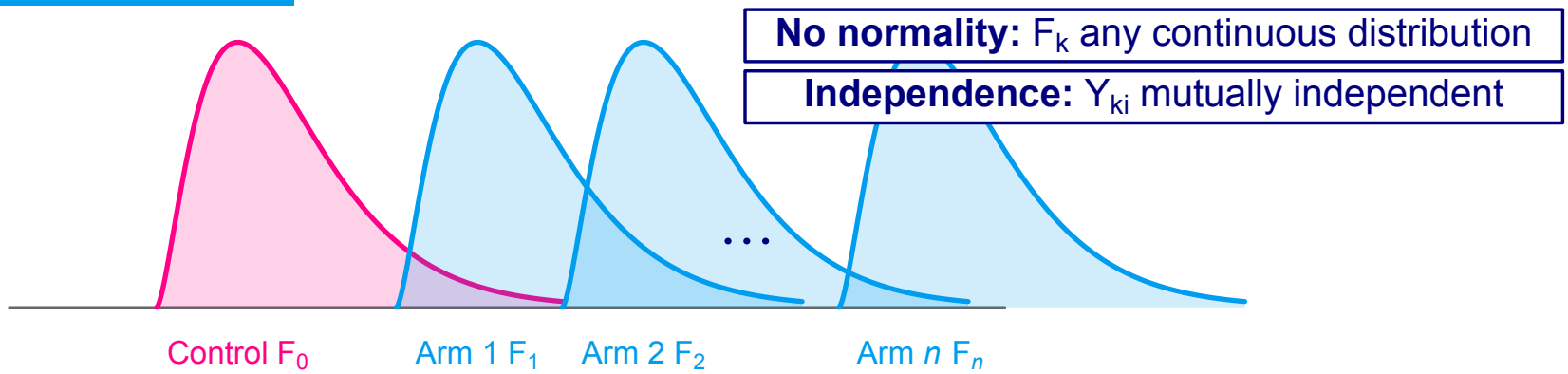
  $\mathcal{D}$ is **approximately** a multivariate Student's t distribution under H_0^{global}

  ↳ **The max-t principle is applied to $\mathcal{D}$**

T2, **T3** and **Games–Howell** treat $\mathcal{D}$ **marginally** (each statistic on its own)

Dunnett-type test without normality (Steel)

- **Statistical model**: one-way layout **without** the normality assumption



- **Hypotheses**: $H_k : F_k = F_0, k = 1, \ldots, n$

  For each of the n arms, the arm distribution equals the control distribution F_0

- Each t_k (standardised Wilcoxon rank sum) is **asymptotically** standard normal under H_k

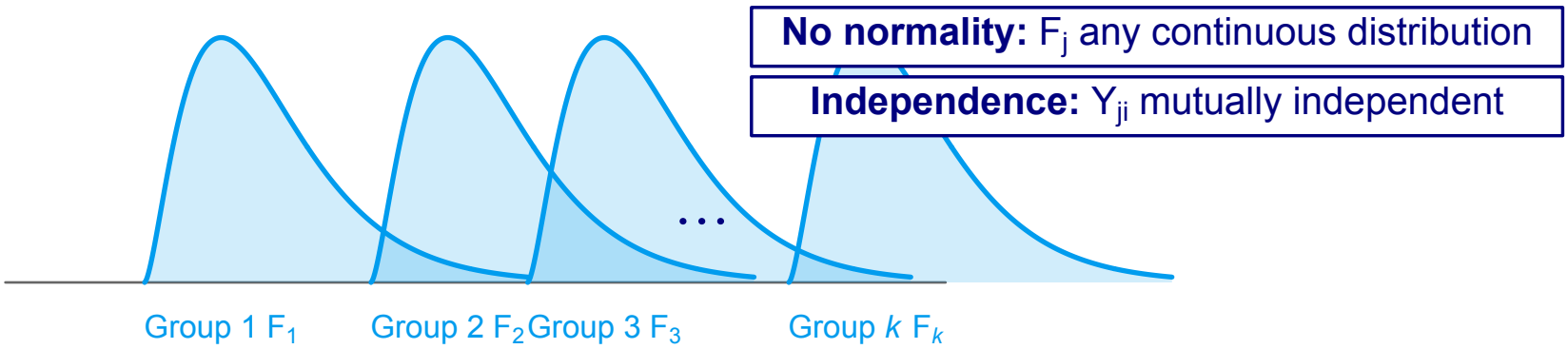
  $\mathcal{D}$ is **asymptotically** a multivariate normal distribution under H_0^{global}

  → **The max-t principle is applied to $\mathcal{D}$**

Šidák treats $\mathcal{D}$ **marginally** (each statistic on its own, not jointly)

Tukey-type test without normality (Steel–Dwass)

- **Statistical model**: one-way layout **without** the normality assumption



- **Hypotheses**: $H_{j\ell} : F_j = F_\ell,\ 1 \le j < \ell \le k$

  For each of the $n = k(k-1)/2$ pairs, the two group distributions are equal

- Each $t_{j\ell}$ (standardised Wilcoxon rank sum) is **asymptotically** standard normal under $H_{j\ell}$

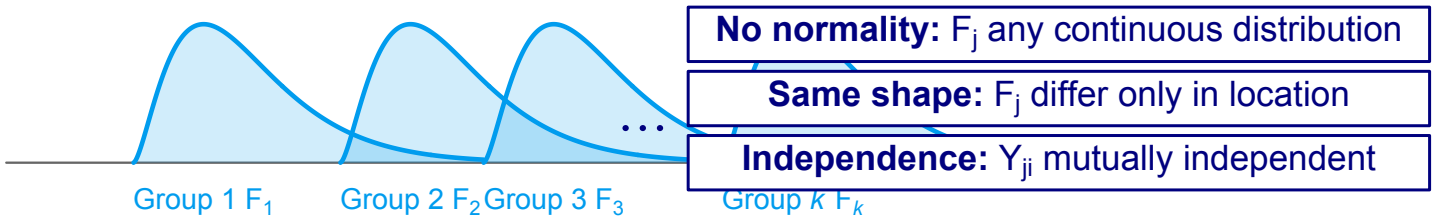
  $\mathcal{D}$ is **asymptotically** a multivariate normal distribution under H_0^{global}

  ↳ **The max-t principle is applied to $\mathcal{D}$**

Šidák treats $\mathcal{D}$ **marginally** (each statistic on its own, not jointly)

Dunn's test (not a max-*t* procedure)

- **Statistical model**: one-way layout **without** the normality assumption



- **Hypotheses**: $H_{j\ell} : F_j = F_\ell, 1 \le j < \ell \le k$

  For each of the $n = k(k-1)/2$ pairs, the two group distributions are equal

- **Test statistic** for each pair, from the **mean ranks** $\bar{R}_j$ over all N observations:

$$t_{j\ell} = \frac{\bar{R}_j - \bar{R}_\ell}{\sqrt{\frac{N(N+1)}{12}\left(1/m_j + 1/m_\ell\right)}}$$

asymptotically $N(0,1)$ under $H_{j\ell}$

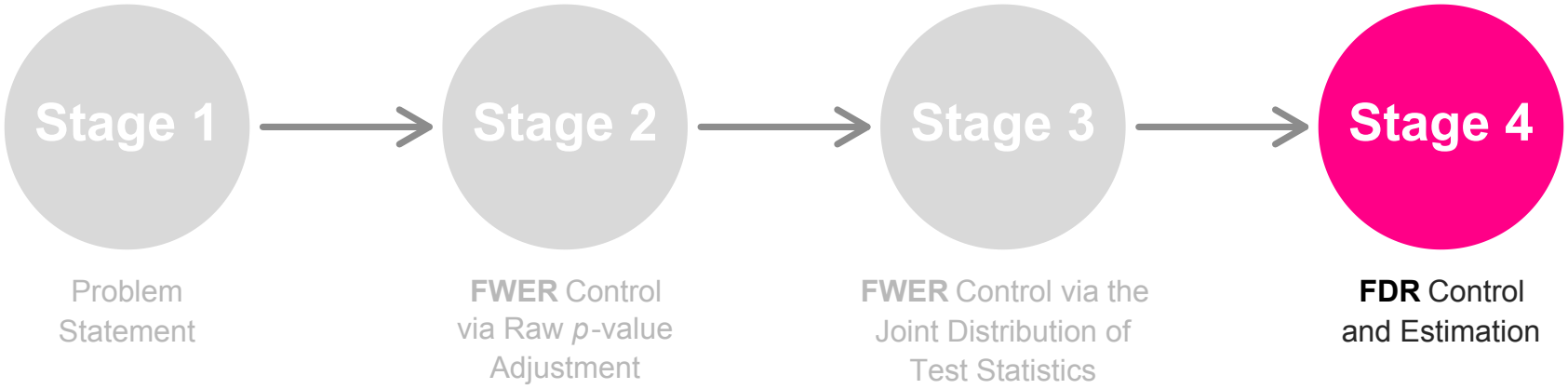
- ***p*-value** of each pair: $p_{j\ell} = 2\left(1 - \Phi\left(|t_{j\ell}|\right)\right)$; **Bonferroni** or **Holm** is then applied to these n p-values. $\mathcal{D}$ is **ignored**

Simulations and practicals

Shiny web applications

https://bioinformatics.cruk.cam.ac.uk/stats/shinystats/

Roadmap

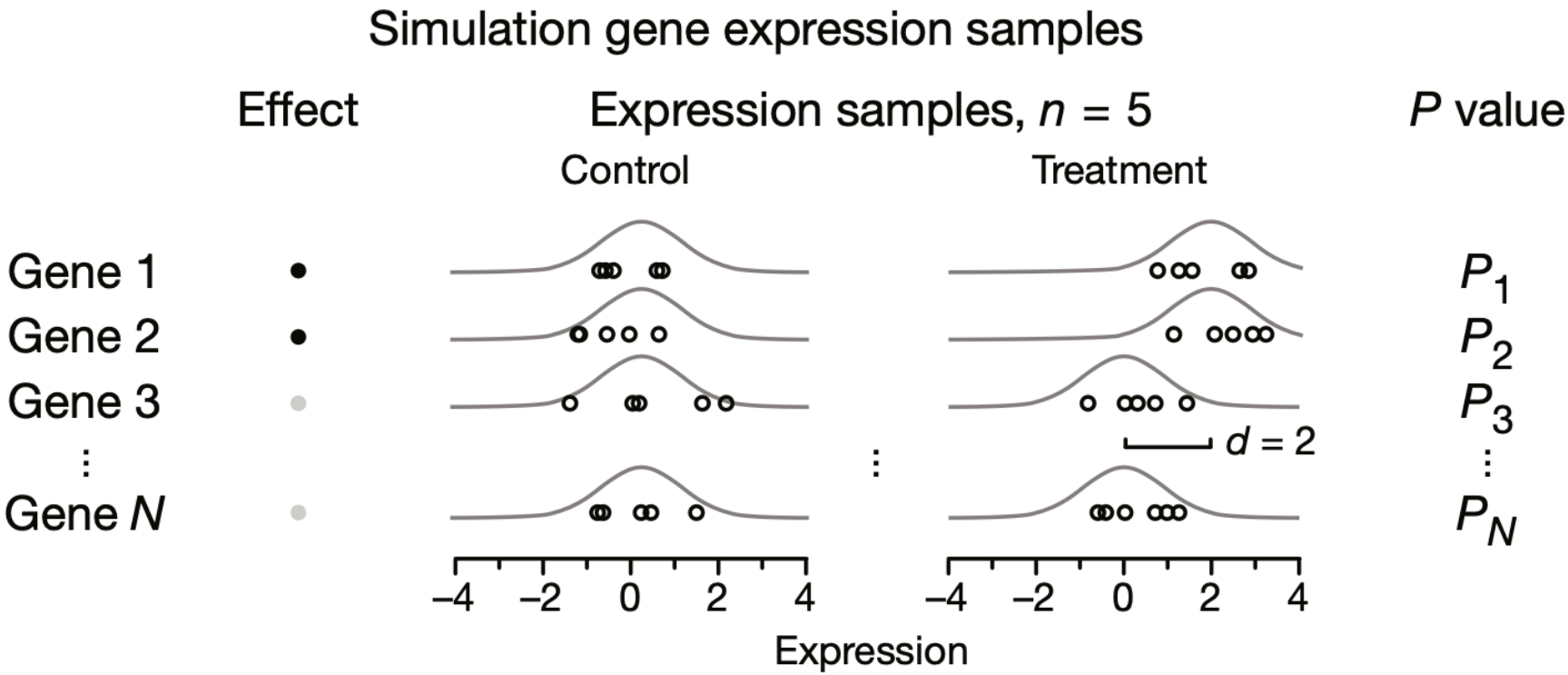


FWER: Family-Wise Error Rate

FDR: False Discovery Rate

FWER and statistical power: the Achilles' heel

For each gene, expression was simulated in a control and a treatment group (5 units each), drawn from a normal distribution ($\mu = 0$, $\sigma = 1$). For a fraction of genes (10% and 50%), an effect size ES $= 2$ ($\sim$ 79% power, $\alpha = 0.05$) was simulated by setting $\mu = 2$ in the treatment sample



Genes were assumed to be statistically independent.

Krzywinski, M. & Altman, N. *Comparing samples — part II*. Nature Methods **11**(4), 355–356 (April 2014).

FWER and statistical power: the Achilles' heel

% of truly differentially expressed genes declared significant

	10% affected			50% affected		
n	Bonf.	Holm	Šidák	Bonf.	Holm	Šidák
10	34.95	35.04	35.37	34.76	37.79	35.15
100	8.70	8.75	8.85	8.72	9.01	8.88
1,000	1.49	1.49	1.52	1.49	1.50	1.52
10,000	0.20	0.20	0.21	0.20	0.20	0.21
100,000	0.02	0.02	0.02	0.02	0.02	0.02

Legend: n = number of genes; Bonf. = Bonferroni correction; 10^5 simulated trials per n

- Power collapses as the family grows: with $n = 10$, about 35% of the truly affected genes are detected; with $n = 100{,}000$, only 0.02%
- All three corrections give essentially the same power. The step-wise gain (Holm) is visible only for small n with many true effects

False discovery rate (FDR)

Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses, and let m_0 ($0 \leq m_0 \leq n$) denote the number of TRUE **null hypotheses**.

$\mathcal{F}$	Not Rejected	Rejected	Total
TRUE	U	V	m_0
FALSE	T	S	$n - m_0$
Total	$n - R$	R	n

- Consider the **proportion** of rejections that are false, $Q = V/R$, with the convention $Q = 0$ when $R = 0$
- Q is a random variable that cannot be observed: R is known once the tests are run, but V is not. We therefore control its **expectation**

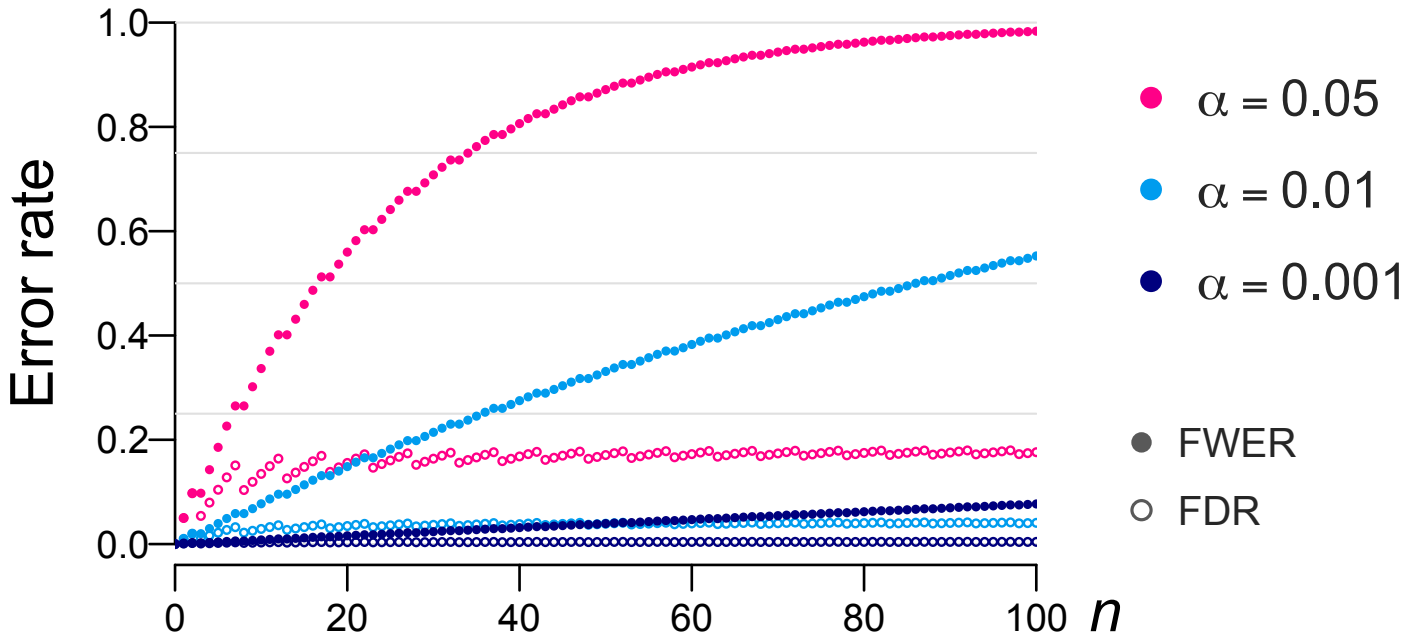
$$\mathbf{FDR} := \mathbf{E}\left[Q \right]$$

We accept that, **on average**, a controlled **proportion of our discoveries is false**.
We tolerate some errors in exchange for more power.

FWER and FDR as the number of tests grows

- Let $\mathcal{F} = \{H_1, H_2, \ldots, H_n\}$ be a finite family of null hypotheses with **independent** test statistics, each tested at level α
- In both series 80% of the nulls are TRUE ($m_0 = 0.8\,n$, no effect); the other 20% are FALSE, each detected with an assumed **power** = **90%** at every α

- **Filled**: FWER $= 1 - (1-\alpha)^{m_0}$ (a probability)
- **Open**: FDR $= \mathrm{E}[V/R]$ (an expected proportion)

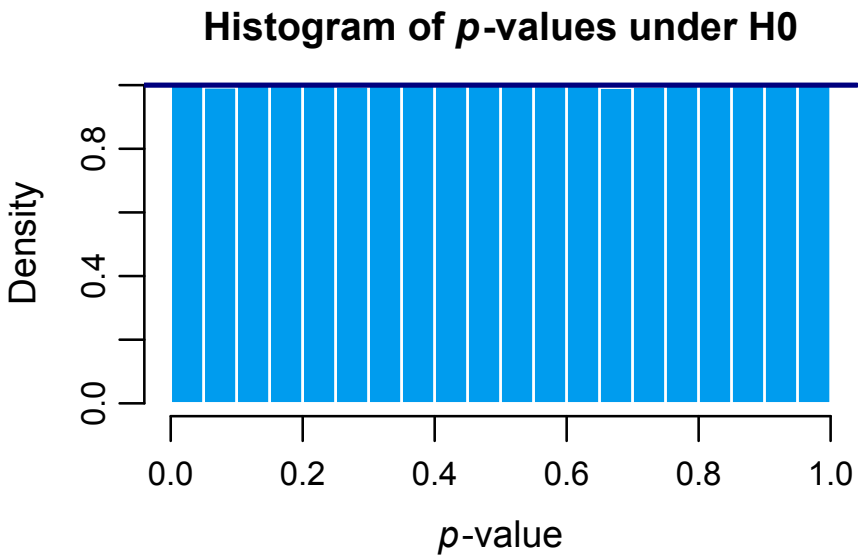


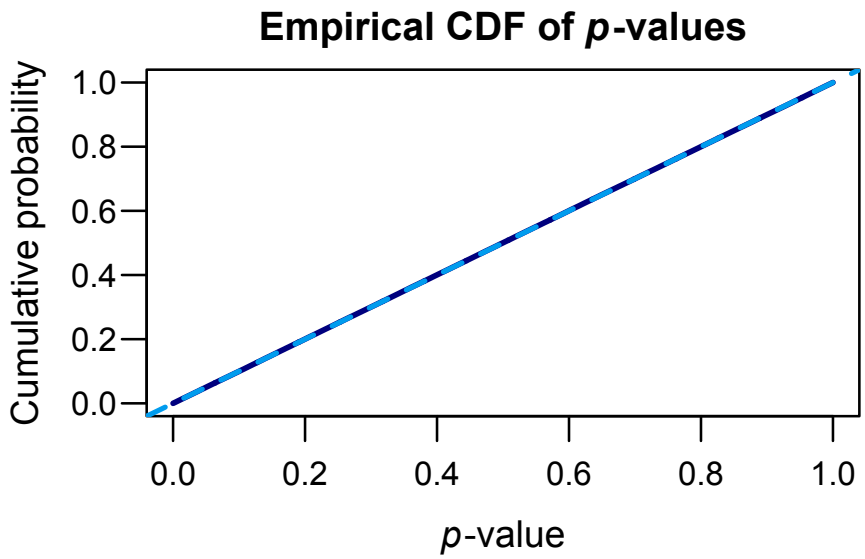
Three questions about the FDR

- **How can we control the FDR?**
- **How can we estimate the FDR?**
- **How can we compute FDR-adjusted *p*-values?**

Uniform distribution of the *p*-value

Let the null hypothesis H_0 be TRUE, and let the *p*-value be computed from the **continuous** null distribution of the **test statistic** *t*. The *p*-value (*p*) is itself a random variable such that $\boldsymbol{p \sim \mathrm{Uniform}(0, 1)}$.





Simulated *p*-values under H_0 ($\mu_1 = \mu_2$) from a two-sample *t*-test with 5 units per group (10^6 simulations): the histogram is flat at density 1 and the empirical cumulative distribution function (CDF) tracks the diagonal $y = x$, confirming $p \sim \mathrm{Uniform}(0, 1)$.

Controlling the FDR: Benjamini–Hochberg (BH)

- **The question**: not *"did I make **any** mistake?"* (FWER), but *"what **share** of my list of discoveries is wrong?"*
- **Count**: reject every $p_i \le \gamma$. Each **TRUE** null is Uniform$(0,1)$, so it falls below γ with probability γ: under H_0^{global} we expect about $n\,\gamma$ **false** discoveries, against the k discoveries we **observe** (k = their number = the rank of the last rejected p-value)
- **Rule**: make that share equal to the FDR level q we are willing to accept (plug-in for FDR $= \mathbb{E}[V/R]$)

$$\frac{\text{expected false}}{\text{observed discoveries}} = \frac{n\,\gamma}{k} = q \qquad \Longrightarrow \qquad \gamma = \frac{\boldsymbol{k\,q}}{\boldsymbol{n}}$$

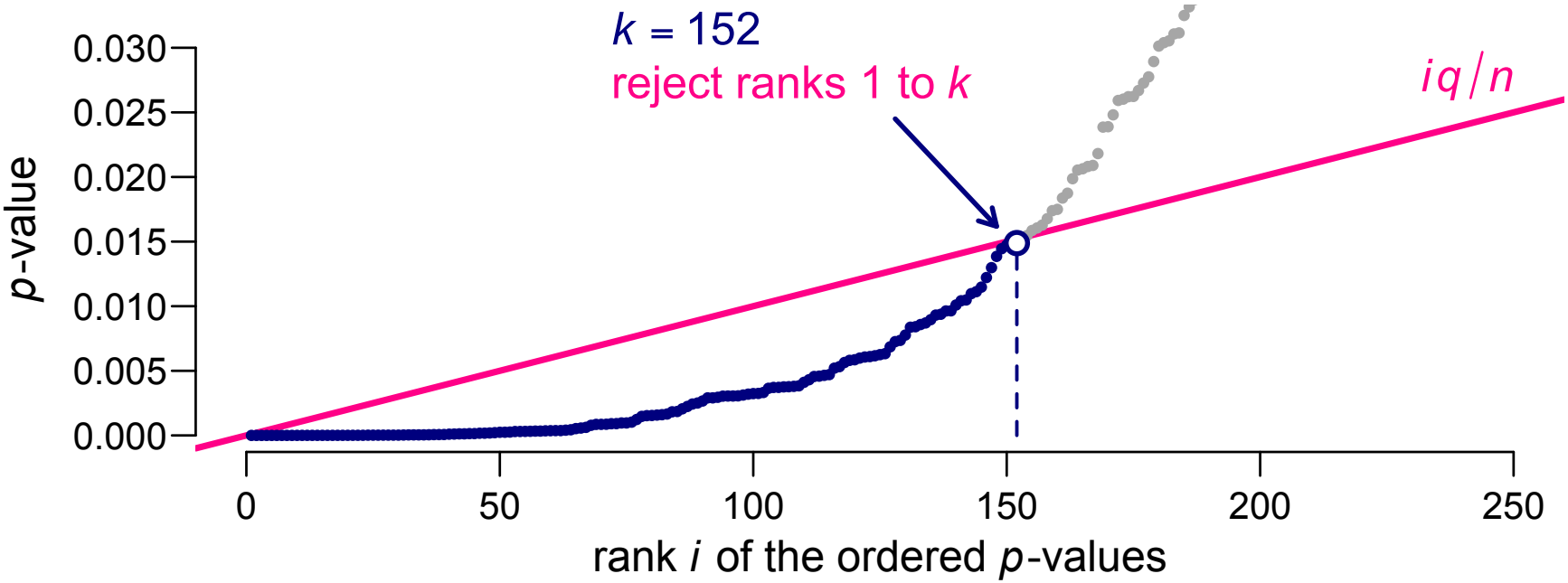
- **Choice**: the threshold γ depends on the rank k, and k depends on γ: BH takes the **largest** k that still satisfies the rule

BH turns the FDR level q into a rank-dependent threshold

Benjamini, Y. & Hochberg, Y. *Controlling the false discovery rate: a practical and powerful approach to multiple testing*. J. R. Stat. Soc. B **57**(1), 289–300 (1995).

Controlling the FDR: Benjamini–Hochberg (BH)

The p-values from $n = 1000$ **independent** tests are **ordered** and compared with the **increasing** threshold $i\,q/n$, to control the **FDR** at $q = 0.10$



rank i	$p_{(i)}$	$i\,q/n$	$\tilde{p}_{(i)}$
152 **(= *k*)**	0.014884	0.015200	**0.09792**
153 **(= *k+1*)**	0.015304	0.015300	0.10003

The adjusted p-value $\tilde{p}_{(i)}$: see the slide *FDR-adjusted p-values: the q-value*

Controlling the FDR: Benjamini–Hochberg (BH)

- Conservative: the rule counts **every** hypothesis as a possible false discovery ($n\,\gamma$), but only the m_0 TRUE nulls can produce one. The real level is therefore

$$\text{FDR} \leq \frac{m_0}{n}\, q \leq q$$

- The factor m_0/n is **power given away**: Storey's $\hat{\pi}_0$ tries to recover it

- Without independence: the proof needs each TRUE null p-value to **move together** with the whole vector of p-values — a small p-value never makes the others **more likely to be large** (*positive regression dependency on a subset*, PRDS). **Positive correlation** is **not always enough**. With **negative** dependence, FDR control is **not guaranteed**

- Solution: **BY**, which divides the threshold by $\sum_{j=1}^{n} 1/j$ and controls the FDR under **any** dependence (here 152 $\rightarrow$ 75 discoveries)

Benjamini, Y. & Yekutieli, D. *The control of the false discovery rate in multiple testing under dependency*. Ann. Statist. **29**(4), 1165–1188 (2001).

Controlling the FDR: Benjamini–Yekutieli (BY)

BH and BY are the **same** step-up rule — order the p-values, take the **largest** k such that

$$p_{(k)} \le \frac{k\,q}{n\,c(n)}$$

BH: $c(n) = 1$

BY: $c(n) = \sum_{j=1}^{n} \frac{1}{j} \approx \ln n + 0.577$

- Assumptions: **none** on the dependence of the p-values
- The price: $c(n) > 1$ makes the threshold $c(n)$ times **stricter at every rank** — not $c(n)$ times fewer discoveries ($c(100) \approx 5.2$, $c(1000) \approx 7.5$, $c(10^4) \approx 9.8$)
- In our $n = 1000$ example, at $q = 0.10$: BH rejects 152 hypotheses, BY only **75**

BY is the dependence-robust version of BH

Controlling the FDR: Storey's procedure

Recover the power BH loses through the factor m_0/n

- BH is **calibrated** for the worst case $m_0 = n$: every hypothesis is a TRUE null. Storey instead takes into account the proportion of true nulls $\pi_0 = m_0/n$
- π_0 is unknown, but it can be **estimated from the data**: $\hat{\pi}_0$
- Apply the **BH rule** with $q/\hat{\pi}_0$ in place of q: take the **largest** k such that

$$p_{(k)} \leq \frac{k\,q}{n\,\hat{\pi}_0}$$ and reject the k smallest p-values

- Since $\hat{\pi}_0 \leq 1$ (it is capped at 1), the threshold is **at least as large** as BH's: **more power**, and the more hypotheses are false ($\pi_0 \ll 1$), the larger the gain
- Assumptions: independence, or only **local** correlation (e.g. small groups of genes), plus a reliable $\hat{\pi}_0$ — which needs **many** tests

How $\hat{\pi}_0$ is estimated: see the slides *Storey's estimate of the FDR – step by step*

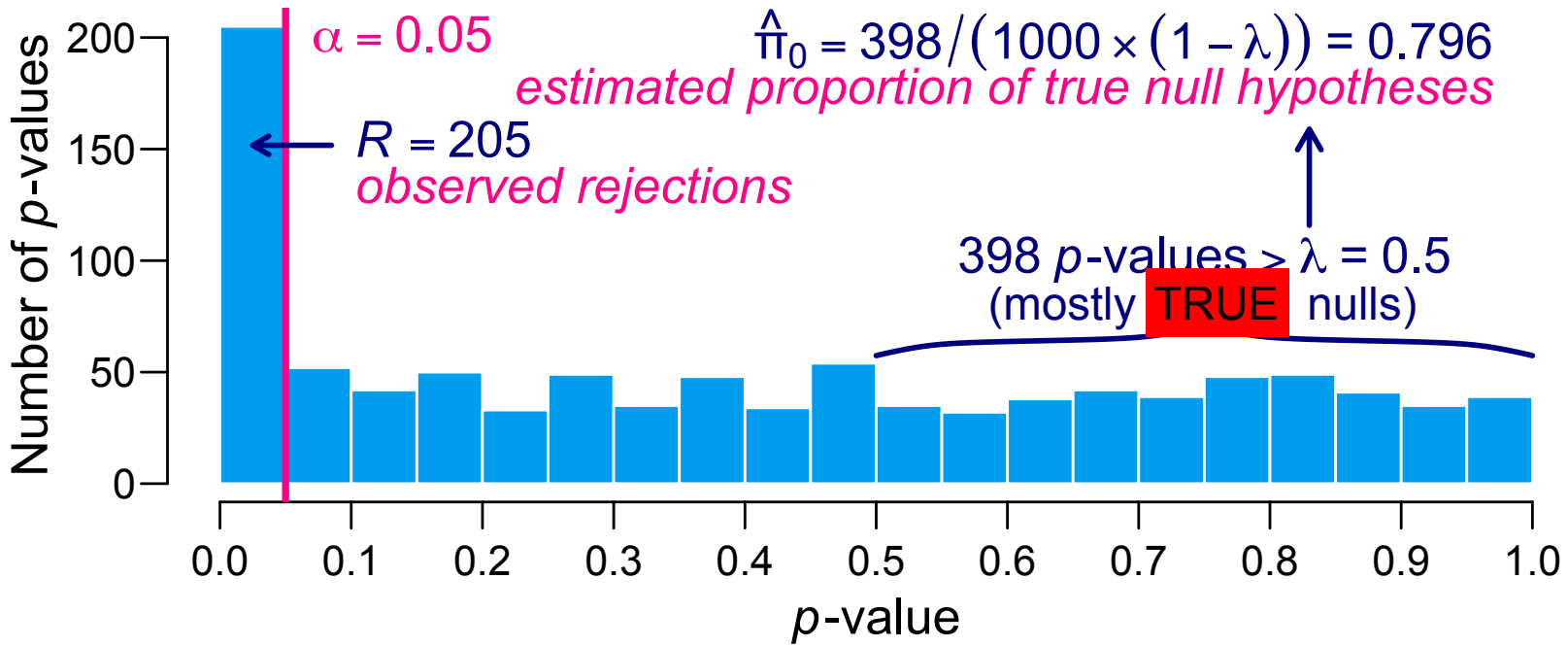
Controlling the FDR: which method?

Method	Assumptions	Power	Best used for
BH	Independence or PRDS	Intermediate	Few true effects ($\pi_0 \approx 1$), or too few tests to estimate π_0
BY	Any dependence	Lowest	Unknown or negative dependence
Storey	Independence or local correlation, plus a reliable $\hat{\pi}_0$	Highest	Many tests, large fraction of true effects ($\pi_0 \ll 1$)

All three order the *p*-values and use a step-up threshold: they differ only in the constant *dividing* $k\,q/n$: fixed (BH, BY) or *estimated* (Storey)

Storey's estimate of the FDR: step by step

The same $n = 1000$ **independent** tests: empirical distribution of the *p*-values, with $\alpha = 0.05$ used as a **fixed cut-off** and $\lambda = 0.5$ above which *p*-values are taken as TRUE nulls



$$\widehat{\mathrm{FDR}}(\alpha) = \frac{\hat{\pi}_0 \times n \times \alpha}{R} = \frac{0.796 \times 1000 \times 0.05}{205} = \mathbf{0.194}$$

Storey, J. D. *A direct approach to false discovery rates*. J. R. Stat. Soc. B **64**(3), 479–498 (2002).

Storey's estimate of the FDR: step by step

- Conservative: $\hat{\pi}_0$ treats every *p*-value above λ as coming from a TRUE null. Some real effects also land there, so $\hat{\pi}_0$ is biased **upwards** and $\widehat{\text{FDR}}$ errs on the **safe** side (too high, never too low on average)
- The bias **shrinks** when power is high or λ is large; but a larger λ leaves fewer *p*-values to count, so the estimate is **noisier** ($\lambda = 0.5$ is the usual compromise)
- Without independence: each null *p*-value is still Uniform$(0, 1)$, but correlated *p*-values move **together**: in a single study their histogram can be far from uniform, so $\hat{\pi}_0$ and $\widehat{\text{FDR}}$ can be far from the truth, **too high or too low**
- Solution: **resample** so that the correlation between tests is kept, e.g. by permuting the sample labels (Storey's permutation method)

Storey, J. D. & Tibshirani, R. *Estimating the positive false discovery rate under dependence, with applications to DNA microarrays*. Technical Report No. 2001-28, Department of Statistics, Stanford University (2001).

FDR-adjusted *p*-values: the *q*-value

- **Definition**: the **smallest FDR** at which a test would still be called significant.
  Note: $q_{(i)}$ is the ***q*-value of one test**, not the FDR **level** q

- **Calculation of *q*-values**: for the ordered *p*-values $p_{(1)} \leq \cdots \leq p_{(n)}$

BH	**BY**	**Storey**
$\min_{j \geq i} \min\left(1, \frac{n}{j}\, p_{(j)}\right)$	$\min_{j \geq i} \min\left(1, c(n)\, \frac{n}{j}\, p_{(j)}\right)$	$\min_{j \geq i} \min\left(1, \hat{\pi}_0\, \frac{n}{j}\, p_{(j)}\right)$

Same three steps: scale $p_{(j)}$ by n/j, take the **running minimum from the right**, cap at 1

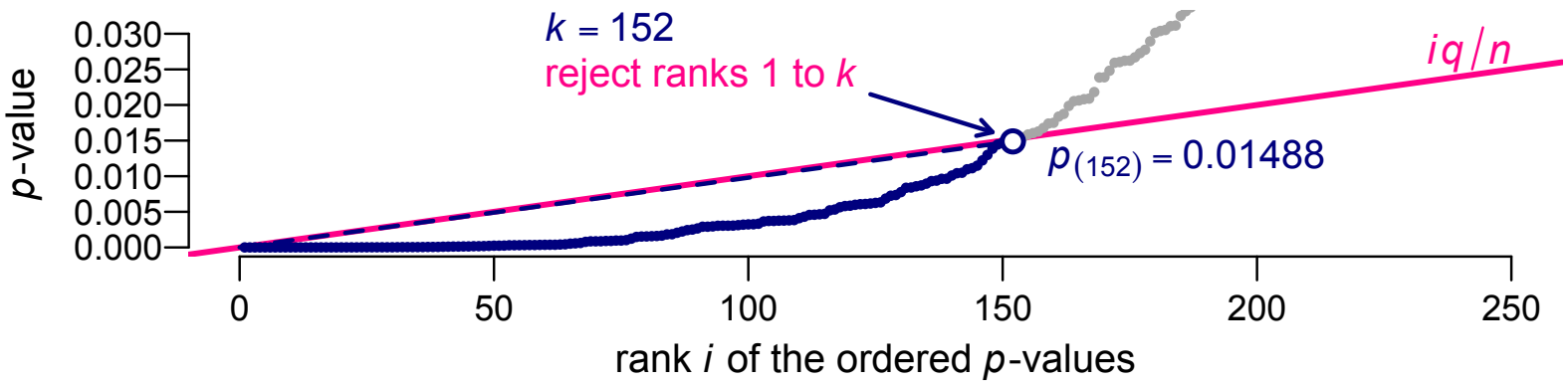
Only the constant in front differs: 1 (BH), $c(n) = \sum_{j=1}^{n} 1/j$ (BY), $\hat{\pi}_0$ (Storey)

- **Interpretation**: reject every test with $q_{(i)} \leq 0.05$ and **at most** $\approx 5\%$ of those discoveries are expected to be false

Storey, J. D. *The positive false discovery rate: a Bayesian interpretation and the q-value*. Ann. Statist. **31**(6), 2013–2035 (2003). Storey, J. D. & Tibshirani, R. *Statistical significance for genomewide studies*. Proc. Natl. Acad. Sci. USA **100**(16), 9440–9445 (2003).

Reading a q-value off the plot

The line from the **origin** through a point has slope $p_{(i)}/i$: it is the BH threshold line at the level $q = n\,p_{(i)}/i$. The **same** point gives **three** adjusted values, one per constant



$$q_{\text{BH}} = \frac{n \times p_{(152)}}{152} = \frac{1000 \times 0.01488}{152} = 0.098$$

here the running minimum is reached at $j = 152$, and nothing is capped at 1

BH	BY	Storey
1×0.098	7.49×0.098	0.796×0.098
0.098	**0.733**	**0.078**

At $q = 0.10$ the same test is rejected by BH and Storey, not by BY

Simulations and practicals

Shiny web applications

https://bioinformatics.cruk.cam.ac.uk/stats/shinystats/